

DSMZ-TCGA Correlation Workflow

This README summarizes the end-to-end workflow for building RNA-seq count data from STAR outputs, performing quality checks, inferring strandedness, and computing DSMZ-TCGA expression correlations with optional tumor purity adjustment and UMAP visualization.

Key Inputs

- FASTQ lists: text file of FASTQ paths for sample discovery.
- BAM files: STAR-aligned, coordinate-sorted BAMs plus .bai.
- STAR ReadsPerGene outputs: one file per sample.
- GENCODE GTF: e.g., Homo_sapiens.GRCh38.107.gtf.
- TCGA SummarizedExperiment: gene counts for TCGA tumors (RDS).
- DSMZ metadata: CSV with sample_name and Origin (organ source).
- (Optional) Purity table: per-sample tumor purity for TCGA.

Key Outputs

- samples.tsv: paired-end sample manifest for downstream use.
- bam_coverage.tsv: mapped-read totals and coverage status.
- *.infer.txt: strandedness inference per BAM (RSeQC).
- DSMZ_star2count.rds: merged counts from STAR outputs.
- DSMZ_count_gene.rds: counts with Ensembl and gene_name columns.
- spearman_all_wide.csv and spearman_all_long.csv: all DSMZ vs. TCGA cohort Spearman correlations.
- dsmz_tcgma_mapping.csv and spearman_best_by_sample.csv: organ-constrained and best overall matches.
- diagnostic PDFs: violin plots, heatmaps, UMAP embeddings.

Workflow Steps

1) Generate Sample Manifest

Script: generate_samples_tsv.py. Reads a text file listing FASTQ files, groups R1/R2 by prefix, and writes config/samples.tsv for downstream processes.

Example:

```
python generate_samples_tsv.py (paths configured in the script)
```

2) BAM Coverage QC

Script: bam_coverage.py. Runs samtools idxstats and totals mapped reads per BAM to flag low-coverage samples.

Example:

```
python bam_coverage.py (adjust bam_dir, output_path, min_reads in the script)
```

3) Strandedness Inference

Script: `run_infer_experiment.py`. Validates BAMs, runs RSeQC `infer_experiment.py` with BED12 annotation, and summarizes strandedness calls.

Example:

```
python run_infer_experiment.py (paths configured at top of script)
```

4) Build DSMZ Count Tables From STAR

R Markdown: `star_2_countdata.Rmd`. Collects all STAR `ReadsPerGene.out.tab` files, extracts unstranded counts, merges across samples, maps Ensembl IDs to gene_name via GENCODE GTF, and writes:

- `DSMZ_star2count.rds` – merged raw counts with `gene_id_ver`.
- `DSMZ_count_gene.rds` – tidy table with `Ensembl_ID_with_version`, `Ensembl_ID` (no version), `gene_name`, then samples.

Example:

```
Rscript -e 'rmarkdown::render("star_2_countdata.Rmd", output_format="pdf_document")'
```

5) Compute Gene Lengths (Optional)

Script: `gtf_exon_lengths.py`. Parses GTF exon features, merges overlaps, and sums non-overlapping exon length per gene.

Example:

```
python gtf_exon_lengths.py (driven by Snakemake or direct args, depending on your variant)
```

6) TPM Calculation (Optional)

Script: `compute_tpm.py`. Merges raw counts with gene lengths and computes TPM per sample.

Example:

```
python compute_tpm.py (driven by Snakemake or direct args, depending on your variant)
```

7) DSMZ-TCGA Correlation Analysis

R script: `dsmz_tcga_correlation.R`. Harmonizes Ensembl IDs (version handling), aligns DSMZ counts to metadata, optionally filters and regresses TCGA by tumor purity, computes TCGA cohort means, selects variable genes if requested, computes Spearman correlations (DSMZ samples by TCGA cohorts), performs organ-constrained mapping, and generates plots (violin, heatmaps) and UMAP embeddings.

Key outputs:

- `spearman_all_wide.csv`, `spearman_all_long.csv` – all correlations.
- `dsmz_tcga_mapping.csv`, `spearman_best_by_sample.csv` – organ-constrained and best overall matches.
- `violin_rho_by_organ.pdf`, `heatmap_top25_dsmz_vs_tcga_cohorts.pdf`, `organ_tcga_assignment_counts.pdf`, `umap_*.pdf` – diagnostics and embeddings.

Example:

```
Rscript dsmz_tcga_correlation.R
```

Notes and Tips

- Keep BAMs coordinate-sorted and indexed. Coverage and strandedness steps assume valid .bam + .bai.
- Ensure genome build consistency between BAMs and BED/GTF (e.g., GRCh38 hg38).
- If you see very few shared genes between DSMZ and TCGA, inspect Ensembl ID versioning and mapping.
- On HPC, submit each step as a standalone job, or wire them with a workflow manager later.